Assessment Task 2: Advanced Data Visualisation-

Australian Open Tennis Championships

Muhammad Hasan Halidh

24749590

Executive Summary

This report explores a dataset consisting of 120 years of Australian Open champions. The data consists of 259 records, across 10 dimensions, including champion names, nationalities, seeds, and set scores. The report will uncover performance trends, gender comparisons, nationality patterns, and top champion statistics/analysis, providing valuable insights for tennis enthusiasts. Excel was used to pre-process and format the data such as using the primary dimensions with unstructured data to create new dimensions that show a clear vision of the scores and performance. Tableau was used to build the insightful visualisations including, treemaps, scatter plots, line graphs, geographic maps, parallel coordinate plots, and bubble charts, to highlight performance patterns and tournament evolution

Dataset Format and Characteristics

Dimension	**Data Type**	**Description**
Year	Quantitative interval	Time intervals are meaningful, and no true zero in the tennis championship timeline.
Gender	Categorical Nominal	This data set contains 2 genders, male and female with no order.
Champion	Categorical Nominal	Names are identifiers, not measurable or ordered.
Champion Nationality/Country	Categorical Nominal	Country names are categories without order.
Champion Seed	Categorical ordinal	Some values are unseeded so the field is treated as a textual dimension. Lower numbers indicate higher-ranked players
Runner-up	Categorical nominal	Names are identifiers, not measurable or ordered.

Runner-up Nationality/Country	Categorical nominal	Country names, no ranking
Runner-up Seed	Categorical ordinal	Some values are unseeded so the field is treated as a textual dimension. Lower numbers indicate higher-ranked players
Score	Textual (String)	Final Set scores which are concatenated into one column
1st won to 5th loss	Quantitative (ratio)	Parsed match set results

Added columns include Champion win count, No. of times this champion has been runner-up, and Winning Rate when reaching the finals, which are all Quantitative (ratio).

Notable Characteristics/trends:

Som visible discrepancies seen in the data include,

- 1 championship games per year for each gender, however in 1977, there were 2 championship games for each gender, resulting in 2 champions and two runner ups per gender.
- Typically, male games are a best of 5 sets, while for women it's a best of 3 sets, reflected in the Score and parsed set columns (1st-won to 5th-loss), where men's matches sometimes span 4–5 sets, while women's usually stop at 2–3.
- Between 1905 to 1923, there was no champion seed or runner up seed, this was because 1905 was when the first Australasian championships started, however only after 1923 was when the International Lawn Tennis Federation decided to "recognise the Australasian Championships on the same footing as those of the United States, England and France", therefore recognised as a major tournament and then establishing seeding for the competitors (Google Arts & Culture, n.d.) . On this note, between 1905 to 1923 there were only male championships, no women.
- There were no championships between 1916 to 1918 due to WW1 and 1941-1945 (WW2 period).

Data Preparation and transformation

The excel file provided for this analysis went under certain transformations and calculations enabling direct comparison of player performance across time, including the identification of the tournament's most successful players. 3 new columns were made:

Champion Win Count:

The number of times each player appeared as a champion was calculated. This column was necessary to isolate players with five or more wins.

=COUNTIF([Champion Column], [Champion])

No. of times this champion has been runner-up Count:

This column counts how many times the champion of the row was a runner-up in the dataset. Again, the COUNTIF function was used, this time on the Runner-up column.

=COUNTIF ([Runner-up Column], [Champion)

This allowed to see the total finals participation, not just wins, and distinguish players who consistently reached the final but won less often.

Winning Rate when reaching the finals

This column calculates the champion of that rows win efficiency, how many finals they won out of their total final appearances.

=[Champion Win Count] / ([Champion Win Count] + [No. of times this champion has been runner-up])

Also, other transformations include, for the jump in data in excel from 1915-1919 and 1941-1945, for better visualisations, new rows were added between those dates with the years listed and the corresponding 'would be' gender for those years, however, no other data present in the other measures/dimensions. This is because Tableau doesn't recognise this as

time jumps, hence not showing consistence in the output of the visualisations. This helped show the time gaps.

e.g

160	1946	Women's	Nancye Wynne Bolton	AUS	Australia	6–4, 6–4	1		6	4	6	4						
161	1946	Men's	John Bromwich	AUS	Australia	5–7, 6–3, 7–5, 3–6, 6–2	1		5	7	6	3	7	5	3	6	6	
162	1945	Women's																
163	1945	Men's																
164	1944	Women's																
165	1944	Men's																
166	1943	Women's																
167	1943	Men's																
168	1942	Women's																
169	1942	Men's																
170	1941	Women's																
171	1941	Men's																
172	1940	Women's	Nancye Wynne Bolton	AUS	Australia	5–7, 6–4, 6–0	1		5	7	6	4	6	0				

Finally, the mins column was also removed, as only 4 data incidents were present out of the entire 226 rows, which cannot be used for any calculations hence regarded as not of any use.

Participation History and Country-Level Dominance

This section explores the overall evolution of the Australian Open from 1905 to 2025, providing insight to the international representation of the champions and the gender population in each nation that has won the AO championships.

Geographic Maps

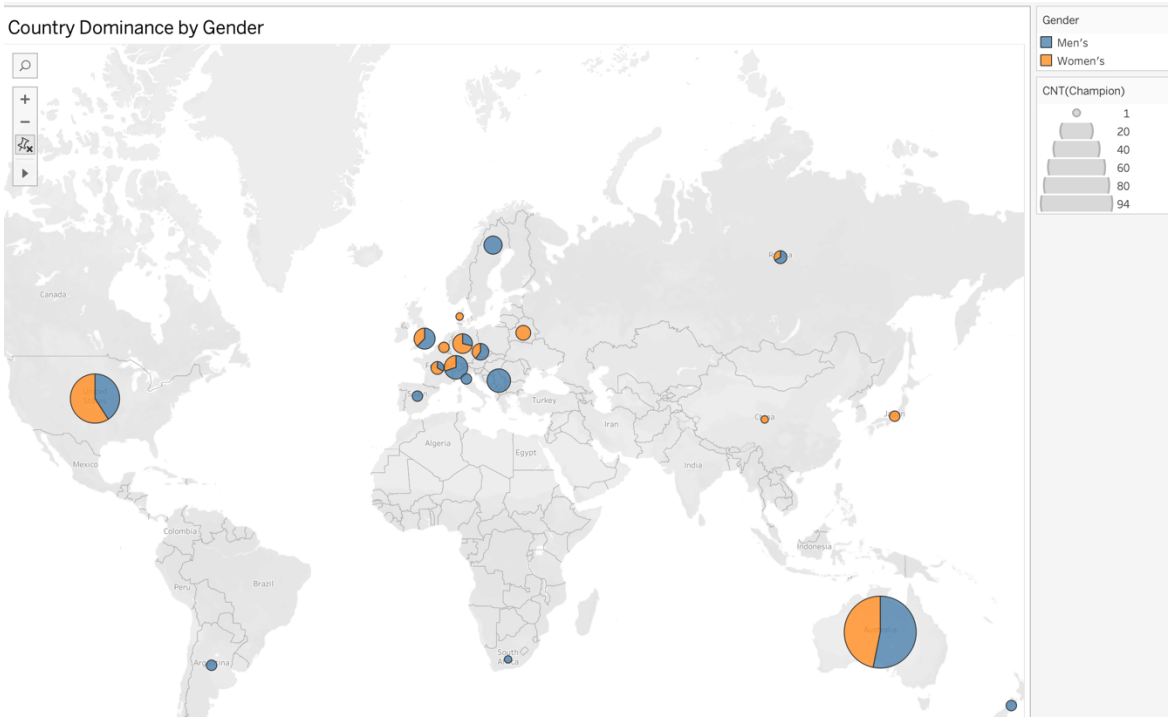

FIgure 1.1 geographic map with pie chart showing country dominance by gender

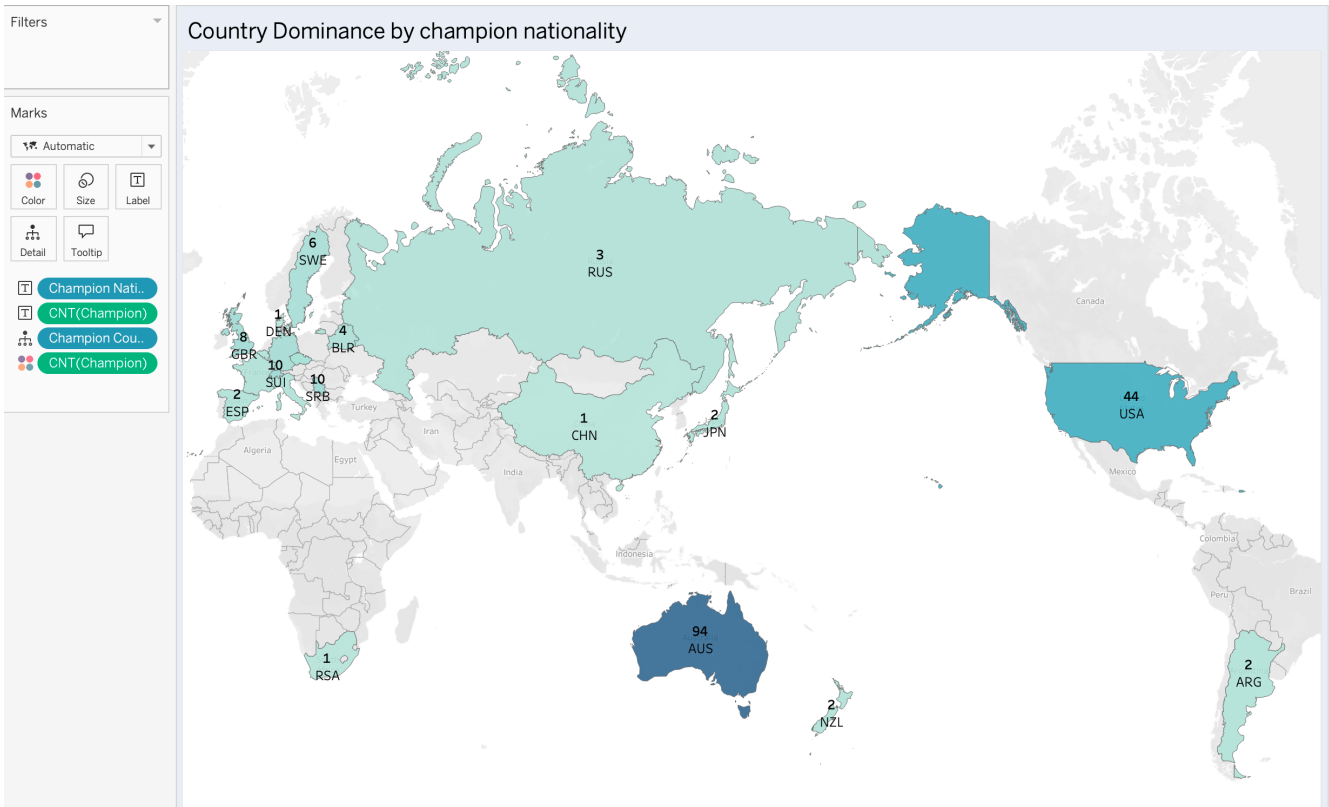

Figure 1.2 showing geographic map with total champions per country.

The map in figure 1.1 displays the total number of championships wins by country. Each country is represented as a pie chart, with the size reflecting

the total number of wins, and colour segments showing the gender ratio (e.g., blue for men, orange for women).

The United States and Australia dominate overall, with strong representation in both genders. Meanwhile, countries like Serbia, Switzerland the UK show male domination, while there are a population of European countries with entirely female domination such as China, Japan, Belarus and Belgium. However this map cannot give quantitative facts as it will cause clutter, hence figure 1.2 aids it to provide a more generalised guide.

Figure 1.2 displays countries by colour, where the darker the blue represents the higher champion count per nation. As seen Australia has the highest count, followed by the USA then the rest of asia/europe.

Bar chart

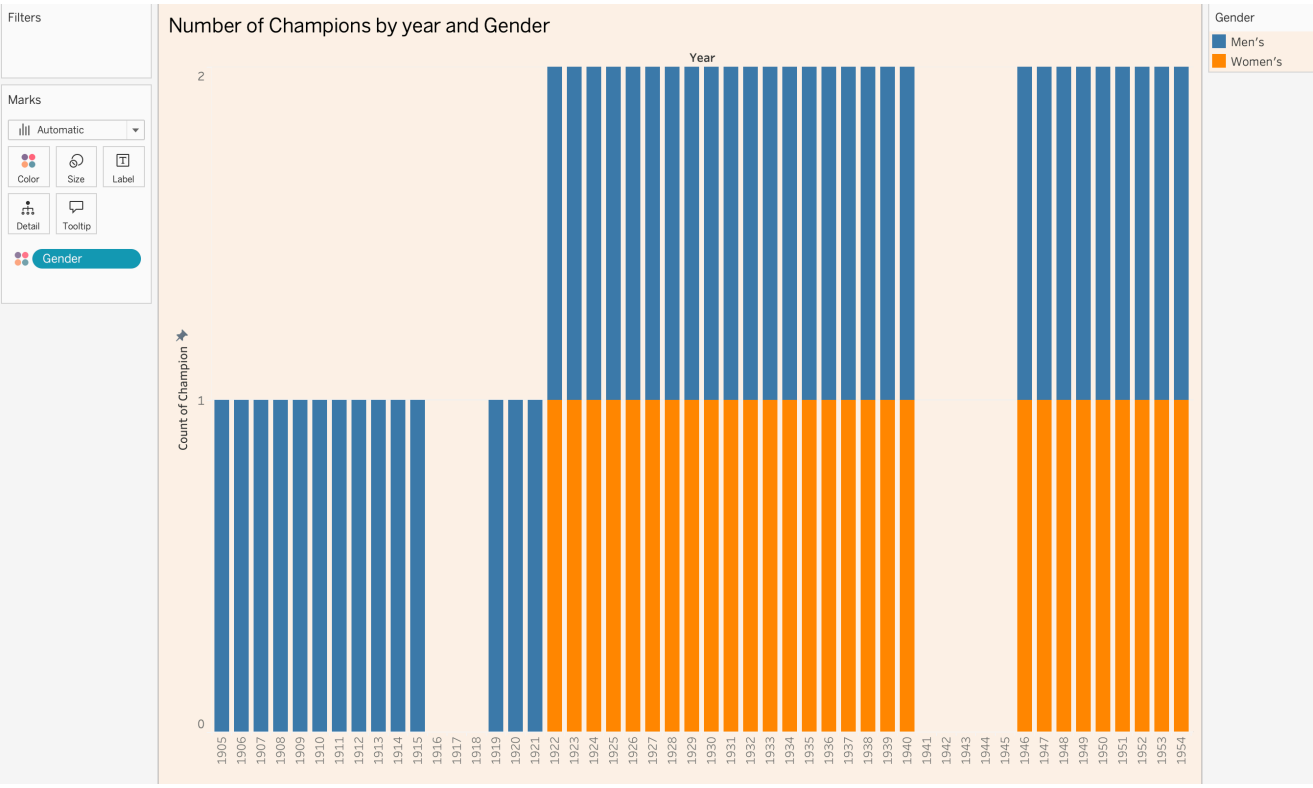

Figure 2 showing time series of number of champions by year and gender

This bar chart shows which years had championship matches, separated by gender. Orange bars indicate women's finals, and blue bars represent men's finals.

Key observations include:

- Clear visual gaps during **World War I (1916–1918)** and **World War II (1941–1945)** when no finals were held.
- Early 1900s show male-only matches; women's finals became established in 1922, which is clearly shown as the orange has overtaken the blue showing that women's was introduced.
- In 1977, two separate tournaments were held, resulting in double bars.

Trends in Champion Seeding

Line chart

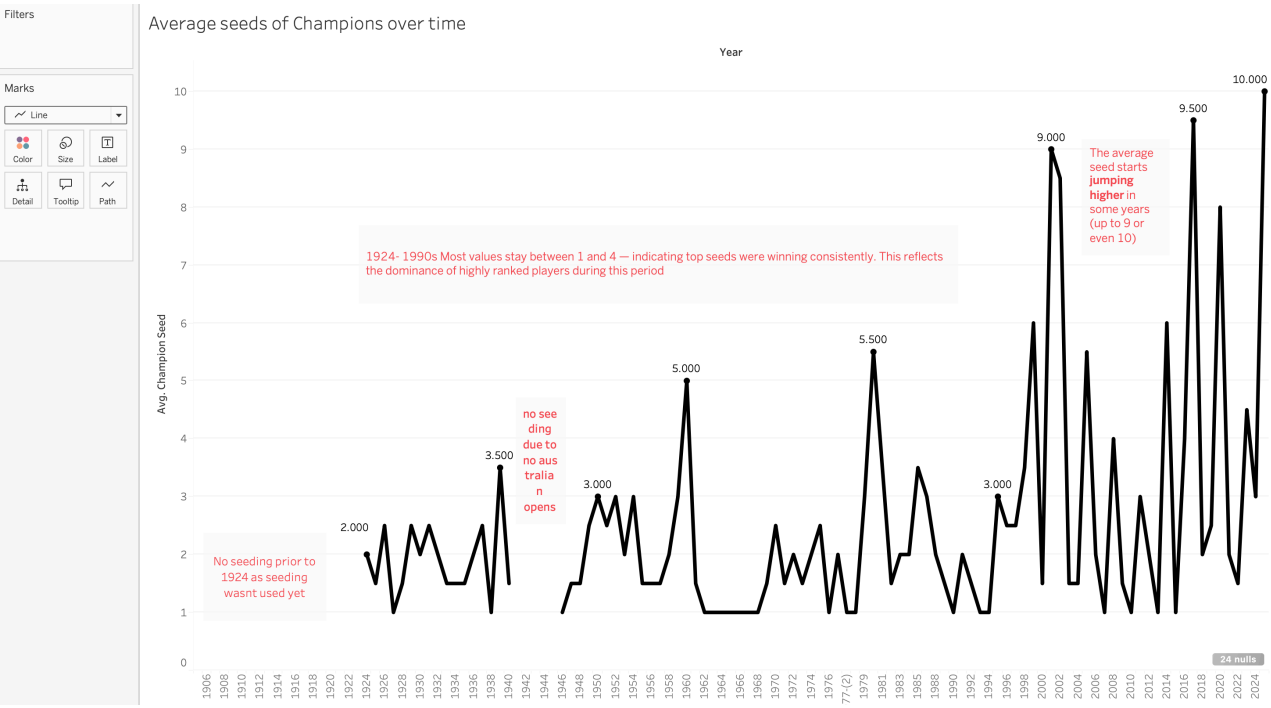

Figure 3 line graph showing the average seed of Champions per year

This line chart illustrates the average seed of Australian Open champions over time. While champions from the 1940s through the 1990s were mostly

top seeds (ranked 1–4), recent decades show more fluctuation, with several years where lower-seeded or unseeded players won. This indicates a growing level of competitiveness and unpredictability. The story telling used in this includes annotations in red above each section describing the scenario at those periods, using area annotations provided in tableau. For example, areas with no seeding is described due to the circumstances present.

Top Australian Open Champions

This section presents a visual comparison of the most successful Australian Open players, defined as those with five or more titles, highlighting their career dominance and performance consistency. The tree map below displays these champions proportionally by win count, with color-coded gender distinctions.

Tree map

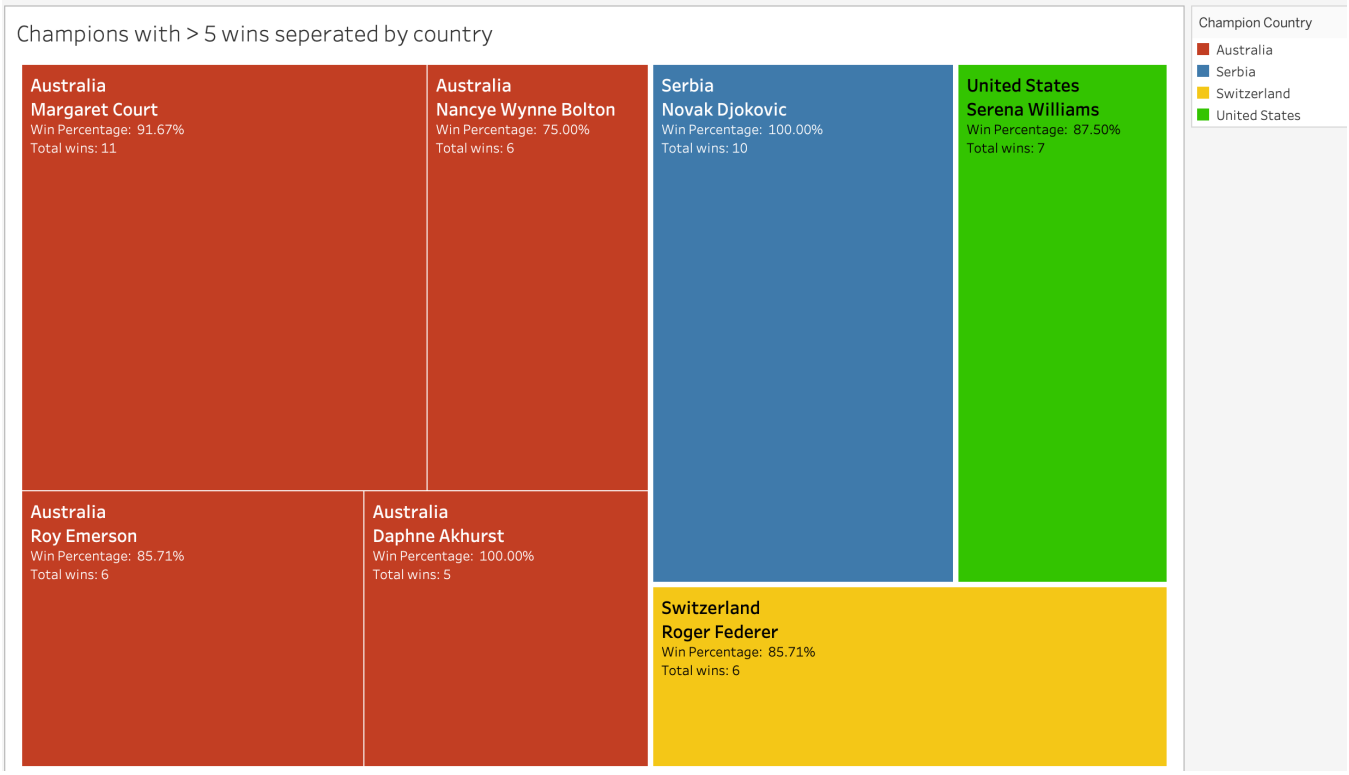

Figure 4 shows treemap of top champions (>5 wins) colour coded by country

The mode of story telling used here is the creation of labels to describe each box in the treemap. Originally, it only consisted of the nationality and name, however, does not provide any further insights into why the treemap is structured this way, hence win percentage and total wins per player was added. Also, the use of colour to distinguish nationalities is used to see the comparison of which countries dominate. As seen above represented in blue, Australia holds 4 top champions accumulating a total of 28 wins. On the right shows the 3 other nationalities including Serbia, Switzerland, and the United States which each have 1 top player, with a total of 23 wins. It seems there was no player from a nationality other than Australia that had an exact number of 5 wins, only 10, 7 and 6 as shown in the treemap above.

Scatter Plot

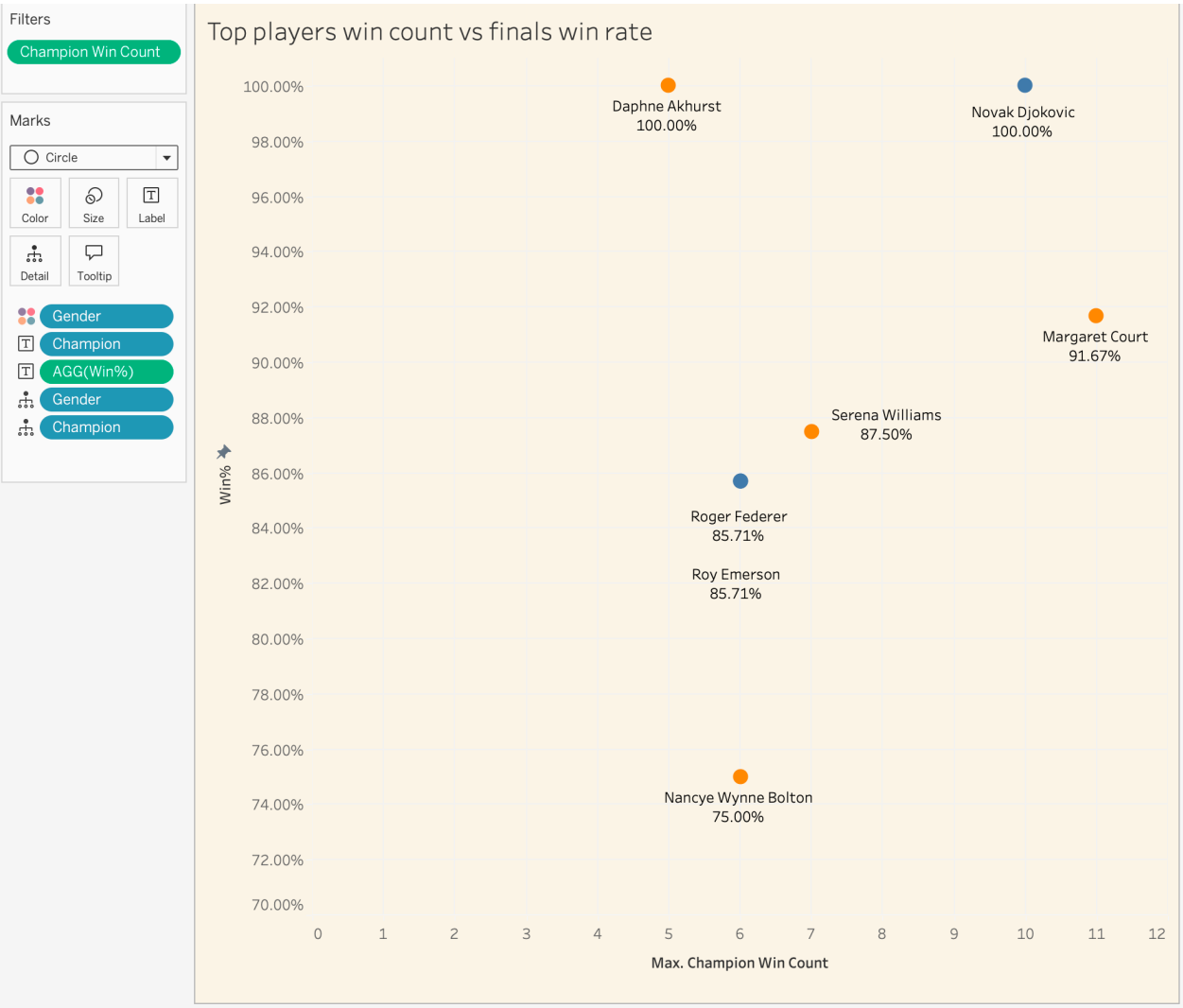

Figure 5 showing a scatter chart of win count of top player (>5 wins) vs win rate.

This scatter plot illustrates how consistent top Australian Open champions have been at converting their final appearances into titles. The x-axis represents the total number of championships won, while the y-axis shows the win rate when reaching the finals. The y-axis is adjusted to start at 70% to avoid clutter, as all featured players have a win rate of 75% or higher. The background is soft and neutral, contrasting the gender-based points, blue for men and orange for women, making it easy to distinguish players. Labels display win rates as percentages for clarity, while win count is easily read from the horizontal axis.

Champion Name	Champion Win Count	No. of times this champion has been runner-up
Margaret Court	11	1
Novak Djokovic	10	0
Serena Williams	7	1
Nancye Wynne Bolton	6	2
Roger Federer	6	1
Roy Emerson	6	1

Figure 6 table displaying the appearance of Champions in finals (refer below)

Novak Djokovic and Daphne Akhurst stand out with a perfect finals win rate of 100%, winning every final they reached. Margaret Court, despite having one runner-up finish, boasts a win rate of 91.67% across 12 finals. Serena Williams follows closely with 87.5%, winning 7 out of 8 finals. Interestingly, Roger Federer and Roy Emerson share the exact same point on the graph, both with 6 wins and a win rate of 85.71%, highlighting identical final records. Nancye Wynne Bolton has the lowest win rate in the group, at 75%, having lost 2 finals out of 8. The clear clustering of men's champions in the upper range of win rate suggests a narrower spread, while women champions display a wider distribution. One possible explanation for this is the best of 5 sets format in the men’s game which may reduce the likelihood of upsets and create a higher competitive nature as noted by The New York Times (Clarey, 2020): “The temptation is to attribute the disparity to the men playing best of five sets and the women best of three, with the shorter format lending itself more easily to upsets.”

Line Chart

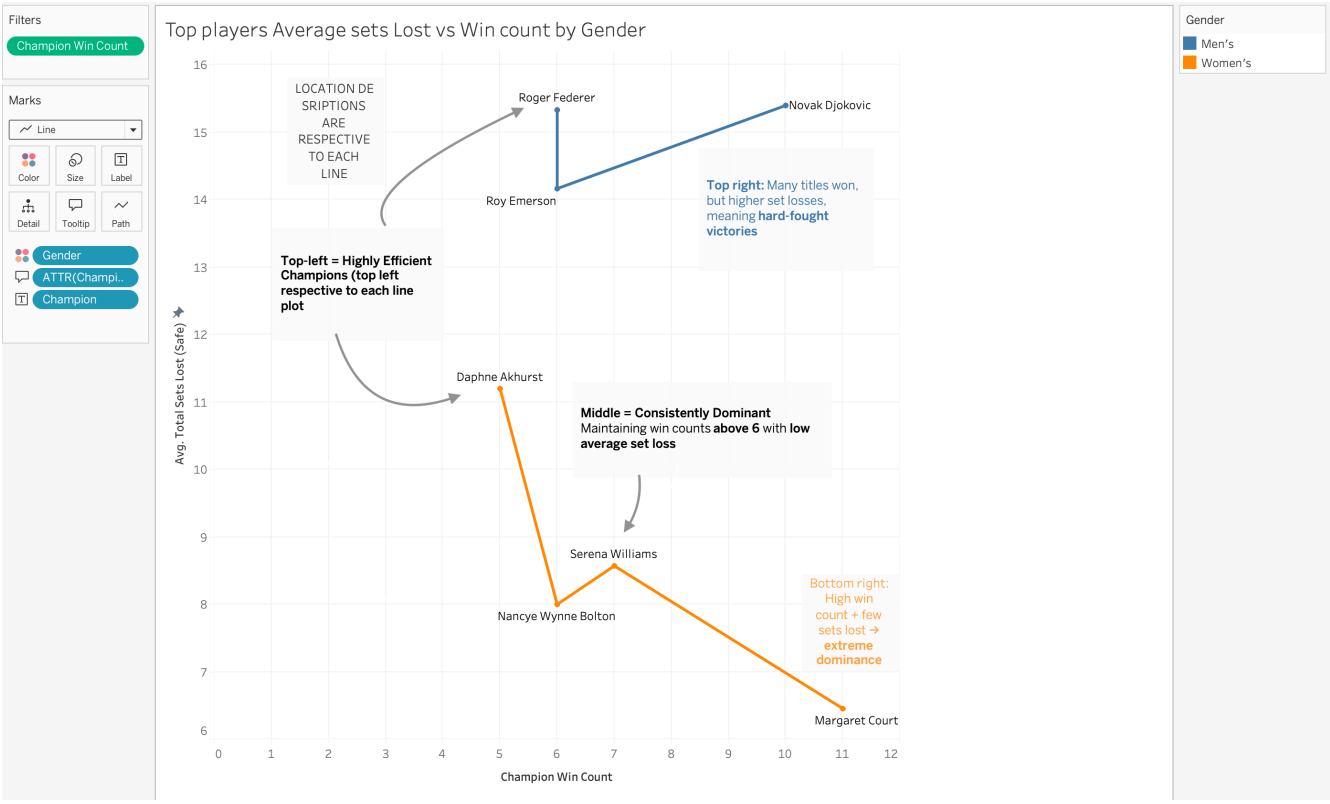

Figure 7 line chart displaying the average sets lost per top player vs their win count

This line plot compares Australian Open champions with five or more titles based on two key performance metrics: total championships won (x-axis) and average sets lost per final (y-axis). Each point represents a player and is colour-coded by gender, enabling quick visual comparison. Text labels next to each point enhance readability and storytelling by directly linking data to individual players. Vectors in the form of arrows are used to link the text labels to their position on the chart. A clear trend emerges: women, who play best-of-3 set finals, appear lower on the graph due to a lower maximum number of sets that can be lost (up to 2), while men, who play best-of-5, naturally average more sets lost (up to 4), placing them higher. For example, Margaret Court stands out in the bottom right, indicating high dominance with 11 titles and very few sets dropped, while Novak Djokovic and Roger Federer appear higher due to the longer format of their matches despite having similarly high title counts. This visual not only compares players by outcomes but reveals deeper differences in performance styles and match structure across eras and genders.

Bubble Chart

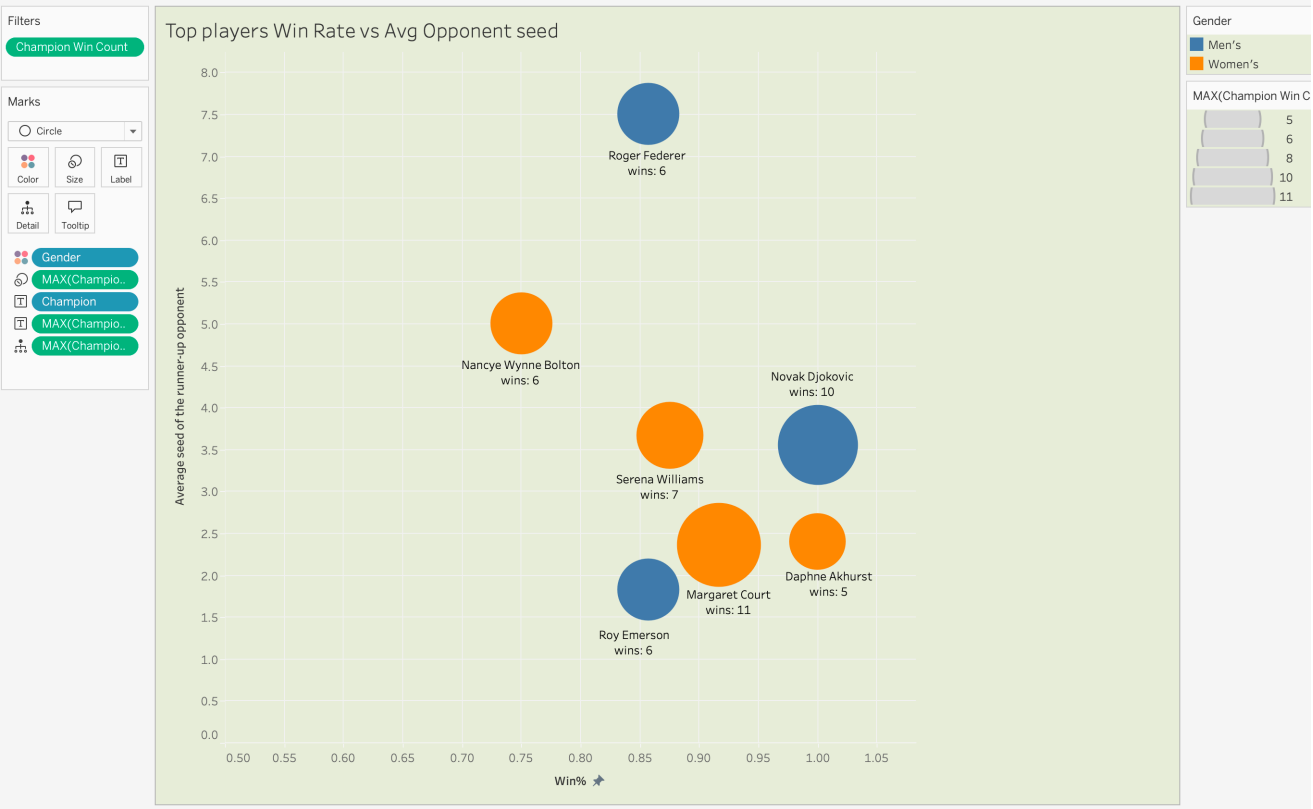

Figure 8 displaying bubble chart comparing win% to average seen of runner up using bubble size to represent total wins per champ

This bubble chart explores the relationship between a champion's final win rate and the average seed of their runner-up opponents, helping to assess the difficulty and efficiency of their victories. Those lower on the chart faced stronger opponents (seeded higher), while those higher up generally faced lower-seeded or unseeded runners-up. For example, Roy Emerson appears in the bottom indicating a near-perfect win record in finals and consistently strong opposition. Novak Djokovic also shows a high win rate with relatively strong opponents. In contrast, Roger Federer appears higher, suggesting he faced weaker opponents on average but had a slightly lower conversion rate. The bubble size communicates each player's historical impact through the number of titles won, while color reveals gender-based comparisons. This visual supports a more nuanced understanding of dominance, highlighting not just who won, but against whom. The colours used display a perfect visualisation which is pleasing to the eye, and the x-axis re-axised to begin from 0.5 as the win% only begins from there.

Parallel Co-ordinate Chart

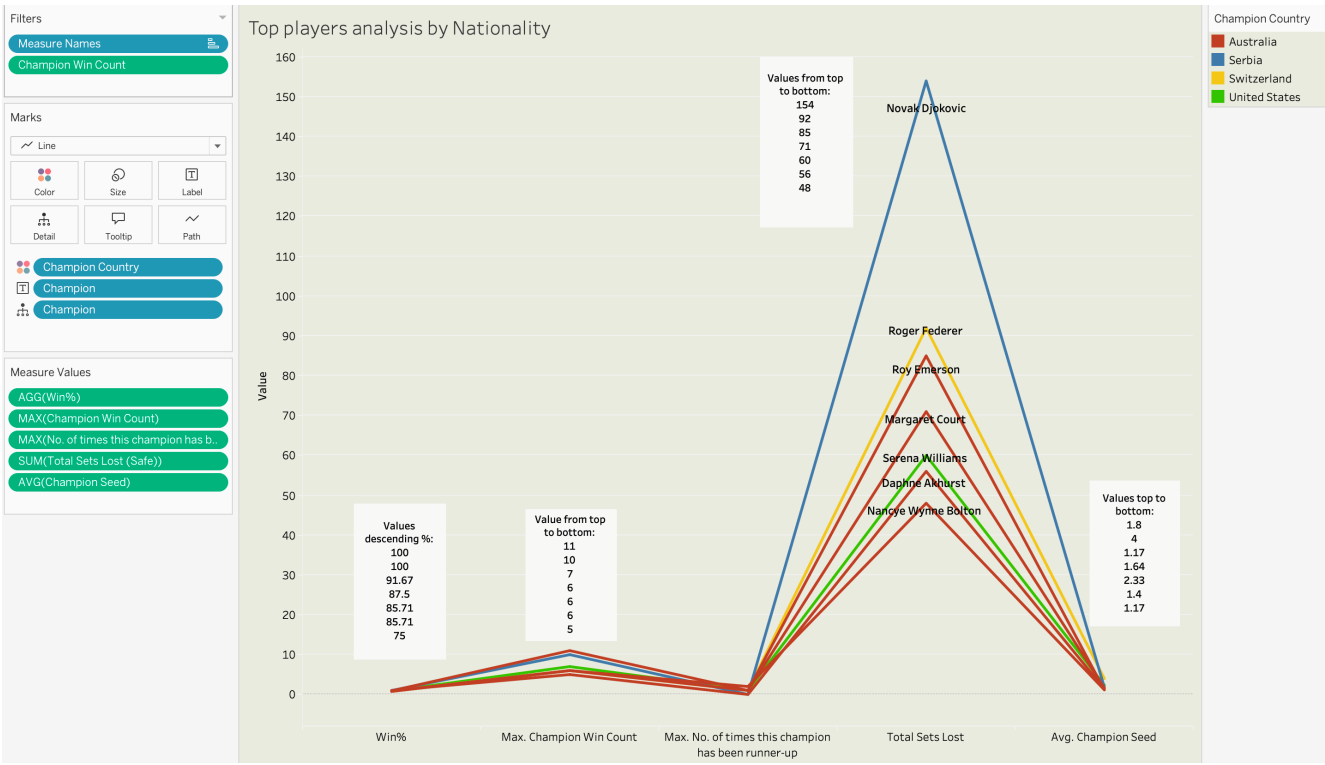

Figure 9 displaying parallel co-ordinate chart comparing multiple metrics for top players coded by nationality

This parallel coordinate plot compares the 7 Australian Open champions who have more than 5 wins, by win percentage, maximum win count, runner-up finishes, total sets lost, and average champion seed, respectively. The order of axes successfully portrays champion efficiency through the order presenting effectiveness, volume, endurance and status. The use of area annotations to clear up cluster and list the data points in order for each necessary dimension allows the viewer to understand and gain a numeric/textual understanding, and the use of contrasting colours to represent nationality on a calm, neutral background allows the easy distinction of nationalities. As seen above, Australian champions Margaret Court and Roy Emerson show strong consistency, while champions such as Novak Djokovic and Roger Federer exhibit variations in their win rates and sets lost. Steep inclines or drops across lines highlight specific strengths or weaknesses for each player. This visualization enables a comparative analysis of top-tier champions while also offering a national context, showing how different countries have historically produced players with varying strengths in dominance, endurance, and ranking.

Conclusion

To conclude, this report provides a comprehensive insight into the Austrlian Open's most successful champions, the 7 individuals who have achieved 5 or more wins, through the use of multi-dimensional visualisations aided by storytelling. Aspects covered were the comparison of player performance across win count, seeding, match outcomes, and opponent strength, highlighting the geographic dominance of champions, and the evolution of the tournament. Gender based comparisons were analysed through the structural match difference between genders and how they impact results. Throughout the report key top players that stood out such as Margaret Court and Novak Djokovic, due to their consistency in winning titles were represented by visualisations illustrating individual excellence and broader tournament trends. Tableau's visualisation tools including treemaps, parallel coordinates, scatter plots, geographic maps, bubble charts, and line/bar graphs enabled a layered, insightful comparison across time and player metrics. Its automated data structuring and simple data transforming system made it easy to transform data on-the go, and the use of calculated fields made it easy to translate raw data into intuitive insights. Also the auto assignation of nationalities into the maps was helpful in organising data. Its interactive capabilities, such as colour coding, area annotations, labelling, made it possible to translate raw data into digestible insights. However, some limitations included the normalisation of axes, not allowing the manual control of scale ranges, reducing the clarity of absolute comparisons. Also, there is difficulty handling categorial data, as Tableau tends to treat mixed-type fields inconsistently. Despite these drawbacks, Tableau remains a powerful platform for visual analysis and effective presentation of multi-dimensional data insights.

Refrences:

1.*Clarey, C. (2020, February 3). The Big Takeaway From Australia: Men's and Women's Tennis Are in Very Different Places. The New York Times. https://www.nytimes.com/2020/02/03/sports/tennis/australian-open.htmlThe Happy*

2.Slam: A History of the Australian Open. (n.d.). Google Arts & Culture. https://artsandculture.google.com/story/the-happy-slam-a-history-of-the-australian-open-tennis-australia/twWhuFJOkd3ELA?hl=enWikipedia Contributors. (2019, November 27). Australian Open. Wikipedia; Wikimedia Foundation.
3.https://en.wikipedia.org/wiki/Australian_Open